

L8: Dynamic Communities – Approach #2

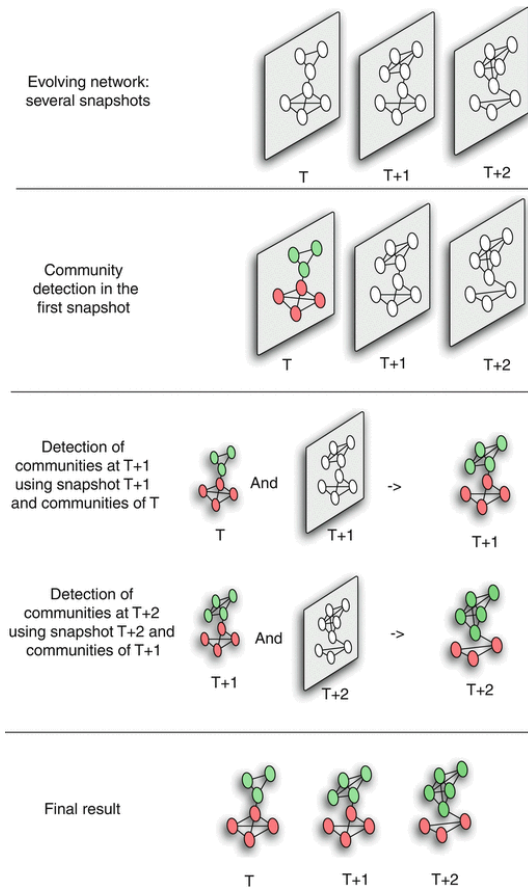


Figure 3 from [Dynamic Community Detection](https://doi.org/10.1007/978-1-4614-6170-8_383) (https://doi.org/10.1007/978-1-4614-6170-8_383) by Cazabet R., Amblard F.

The second approach attempts to compute a good community structure at time T+1 also considering the community structure computed earlier at time T.

To do so, we need to modify the objective function of the community detection optimization problem: instead of trying to maximize modularity at time T+1, we aim to meet two goals simultaneously: have both high modularity at time T+1 and maintain the community structure of time T as much as possible.

This is a trade-off between the quality of the discovered communities (*quantified with the modularity metric*) and the **"smoothness"** of the community evolution over time.

Such dual objectives require additional optimization parameters and trade-offs, raising concerns about the robustness of the results.

Food for Thought

Write down a mathematical expression for a dual optimization objective mentioned above. You know how modularity is defined. How would you define the smoothness of community evolution?